

● *Original Contribution***TARGETED CONTRAST-ENHANCED ULTRASOUND IMAGING OF ANGIOGENESIS IN AN ORTHOTOPIC MOUSE TUMOR MODEL OF RENAL CARCINOMA**SHUPING WEI,^{*} NINGHUA FU,^{*} YU SUN,[†] ZHIJIAN YANG,[†] LI LEI,^{*} PENGFEI HUANG,^{*} and BIN YANG^{*}^{*}Department of Ultrasound Diagnostics, Jinling Hospital, Nanjing University School of Medicine, Nanjing, Jiangsu, China; and[†]Department of Pharmacological Study, Origin Biosciences, Nanjing, Jiangsu, China

(Received 7 April 2013; revised 17 November 2013; in final form 3 December 2013)

Abstract—Previous studies have reported that microbubbles bearing targeting ligands to molecular markers of angiogenesis can be successfully detected by ultrasound imaging in various animal models of solid cancer. In the present study, we sought to investigate the activity of microbubbles targeted to vascular endothelial growth factor receptor 2 (VEGFR2) in an orthotopic model of renal cell carcinoma (RCC). Microbubbles conjugated to an anti-VEGFR2 antibody (MB_V) were compared with microbubbles conjugated to an isotype control antibody (MB_C) or naked microbubbles (MB_N). An orthotopic mouse model of human RCC was established by surgically implanting an established tumor within the renal capsule in mice. Tumor growth and blood flow were verified by B-mode and color Doppler ultrasound imaging. VEGFR2 expression within the tumor and renal parenchyma was detected by immunohistochemistry. The duration of contrast enhancement of MB_V was much longer than those of MB_N and MB_C when assessed over 10 min. The baseline-subtracted contrast intensity within the tumor was higher for MB_V than for MB_C and MB_N ($p < 0.01$). Additionally, the contrast intensity for MB_V was significantly higher in the tumor region than in normal parenchyma ($p < 0.01$). Microbubbles targeting VEGFR2 exhibit suitable properties for imaging angiogenesis in orthotopic models of renal cell carcinoma, with potential applications in life science research and clinical medicine. (E-mail: yb12yx@hotmail.com) © 2014 World Federation for Ultrasound in Medicine & Biology.

Key Words: Targeted ultrasound, Microbubbles, Contrast agent, Molecular imaging, Contrast pulse sequence, Orthotopic, Renal clear cell carcinoma, Nude mouse, Angiogenesis, Vascular endothelial growth factor receptor 2.

INTRODUCTION

It is widely accepted that angiogenesis, the process of new blood vessel formation and recruitment, is critically involved in the early progression and metastatic potential of many types of cancer (Carmeliet 2005; Hicklin and Ellis 2005). Various specific molecular markers of angiogenesis have been identified that are overexpressed on tumor endothelial cells, including vascular endothelial growth factor receptor 2 (VEGFR2). VEGFR2 is a receptor tyrosine kinase that mediates most of the pro-angiogenic activity of VEGF (Hicklin and Ellis 2005; Shibuya 2006; Vajkoczy et al. 2002). VEGFR2 expression within the tumor vasculature has been linked to the progression

and aggressiveness of many types of tumors, including renal cell carcinoma (RCC) (Hicklin and Ellis 2005; Rini et al. 2011).

Renal cell carcinoma accounts for 2%–3% of all adult malignancies and is the third most frequent urologic malignancy after prostate and bladder cancer (American Cancer Society 2008; Rini et al. 2009). Anti-angiogenic drugs targeting the VEGF/VEGFR2 pathway have found particular utility in advanced RCC; these include inhibitors of VEGF receptor such as sorafenib and sunitinib (Escudier et al. 2007; Motzer et al. 2007; Terakawa et al. 2011). However, individual response rates vary among patients, and patient selection to identify responders before implementing treatment is a critical unmet need. Additionally, non-invasive and reproducible methods to assess response to therapy are needed. Finally, quantitative tools for assessing tumor function and phenotype are becoming increasingly important in the context of basic science and the drug discovery process. We

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hypothesize that ultrasound-based molecular imaging may be useful for each of these applications.

Molecular imaging technology has the potential to detect molecular transformations in various lesions non-invasively and in advance of traditional biopsy-based methods. With the development of molecular imaging, contrast-enhanced ultrasound (CEUS) with microbubbles targeting specific molecular markers is emerging as a uniquely effective imaging modality because of its tolerability, low cost, portability and ability to perform real-time anatomic imaging (Klibanov 2006). By conjugation of targeting ligands (such as antibodies, peptides or small molecules) to its surface, the microbubble can be made to target a desired molecular marker of disease. Subsequent ultrasound imaging enables quantitative detection of the target-bound microbubbles. This is inherently a “late-phase” technique, in which scanning occurs 5–15 min after microbubble administration to ensure that circulating (non-target bound) microbubbles have cleared the imaged tissue. Microbubbles are purely intravascular agents, which makes them uniquely suited for detecting and tracking biological processes on vascular endothelial cells, such as tumor angiogenesis. As a key player in the physiology of angiogenesis and as a drug target, VEGFR2 is a potentially powerful target for molecular imaging.

This feasibility of targeting VEGFR2 with targeted microbubbles has been reported in numerous preclinical studies in mice and rats (Anderson *et al.* 2010, 2011; Rychak *et al.* 2007; Willmann *et al.* 2008, 2010). However, to translate this technology into routine use, further information regarding its behavior must be obtained. In the present study, we have chosen to evaluate a commercially available VEGFR2-targeted agent in an orthotopic model of renal cell carcinoma. This allowed us to evaluate whether VEGFR2 is a suitable imaging target in a model that closely mimics naturally occurring RCC. The expression patterns, growth characteristics, vascular arrangement and geometry of orthotopically grown tumors are known to differ substantially from those of subcutaneous models, and it is important to establish that VEGFR2-targeted microbubbles can bind to these tumors in quantities sufficient for imaging. Additionally, locating a tumor mass within a highly vascularized organ such as the kidney presents some degree of technical challenge. Finally, the time course of microbubble wash-in and uptake is unknown. Our goal in this study was to evaluate the time course and specificity of VEGFR2-targeted microbubbles in an orthotopic model of RCC using non-invasive ultrasound molecular imaging.

METHODS

Animal and cancer cell line

Fifteen 6-wk-old male BALB/C nude mice weighing 18–25 g were purchased from the Beijing Kelihua

laboratory animal center (Beijing, China). Animals were maintained in a HEPA-filtered environment, where the room temperature was 24°–25°C and the humidity was 50%–60%. The animals were fed an autoclaved laboratory rodent diet. The protocol for animal experiments was approved by the institutional animal care and use committee at our institution. Human renal clear cell carcinoma cells (line 786-0) were purchased from the Type Culture Collection of the Chinese Academy of Science (Shanghai, China). The cells were maintained in RPMI 1640 (GIBCO Life Technologies, New York, NY, USA) supplemented with 10% heat-inactivated fetal bovine serum (FBS, Hyclone, Logan, UT) and incubated at 37°C in 5% CO₂.

Orthotopic mouse tumor model

We harvested the 786-0 cells at 80% confluence by digestion with 0.25% trypsin solution, washed them twice with phosphate buffered saline and resuspended them at a concentration of 5×10^7 /mL. Approximately 5×10^6 cells (0.1 mL) of the cell suspension were injected subcutaneously into the right flank region of each nude mouse. Tumors grown in nude mice were harvested at the exponential growth phase and resected under aseptic conditions. Necrotic tissues were removed and viable tissues were cut with scissors and minced into 1-mm³ pieces. For surgical orthotopic implantation, animals were anesthetized by injection of 0.02 mL of solution of 50% ketamine, 38% xylazine and 12% acepromazine maleate. A 1-cm vertical incision was made in the skin above the right kidney region using surgical scissors. The kidney was exposed and a small cut made on the renal subcapsule. One piece of tumor fragment was inserted into the capsule, and the cut was then covered with surrounding soft tissue using an 8-0 sutures. The kidney was then returned to the peritoneal cavity, and the incision was closed in layers using sterile 5-0 surgical sutures. All surgical procedures and animal manipulations were conducted under aseptic conditions in an HEPA-filtered laminar-flow hood with a $\times 8$ surgical microscope (YZ20P5, Shanghai, China).

Targeted contrast agent preparation

Streptavidin-coated microbubbles (Targestar-SA) were obtained from Targeson (San Diego, CA, USA; distributed by Origin Biosciences in China). Biotinylated rat anti-mouse VEGFR2 monoclonal antibody (clone Avas12a1, 13-5821, eBioscience, San Diego, CA, USA) was used as the targeting ligand (Rychak *et al.* 2007). Biotinylated isotype-matched rat control immunoglobulin G (IgG) antibody (13-4321, eBioscience) was used as a specificity control. Targestar-SA agents are streptavidin-coated microbubbles composed of a perfluorocarbon gas core encapsulated by a lipid shell. The outer

shell of the microbubble is coated with streptavidin immobilized on a polymer tether, to which the biotinylated ligands were conjugated. Targestar-SA was conjugated with anti-VEGFR-2 antibody (MB_V) and control IgG antibody (MB_C) according to the manufacturer's instructions and freshly prepared before use. Briefly, 100 μ g of biotinylated antibody was injected into the Targestar-SA vial and incubated at room temperature for about 20 min with gentle shaking to keep the agents from caking. Unconjugated ligand was removed by centrifugal washing. The washing procedure results in a slight reduction of the stock microbubble concentration (final microbubble concentration of $\sim 1.5 \times 10^9$ /mL), but did not alter the size distribution. Unconjugated Targestar-SA microbubbles (MB_N) were used as a second negative control.

Targeted contrast-enhanced ultrasound imaging

The mice were anesthetized and then placed in a supine position on a heated stage. Ultrasound imaging was performed using a Siemens Sequoia 512 ultrasound scanner with a 15L-8 linear transducer (Siemens, Mountain View, CA, USA) using contrast pulse sequence (CPS) imaging mode and a transmission frequency of 7 MHz. The transmitted power for CPS pulses was -23 dB, which corresponded to a low mechanical index (MI) of 0.14. The imaging frame rate was set at 16 Hz, CPS gain was set at 0 and the dynamic range was set at 65 dB. The transducer was coupled to the mouse skin in the tumor region with acoustic coupling agent. The ultrasound probe and the animal were both fixed and remained at the same position throughout the study. The acoustic focal zone was placed at the center of the tumor at the largest transverse cross section, and the imaging gain settings were optimized and maintained throughout each experiment.

A conventional B-mode ultrasound scan without contrast agents was first conducted to obtain the location, size and baseline echogenicity of the tumor. Color Doppler flow imaging was then performed to visualize blood flow in the tumor. The scanner was placed in CPS mode, and contrast settings were optimized. A 10- μ L volume of MB_N, MB_C or MB_V (approximately 1×10^7 microbubbles) was intravenously administered by a bolus retro-orbital injection. Each mouse was administered all three microbubble preparations in random order, at intervals of 30 min between injections to allow clearance of previously injected MBs. Full clearance of each dose from the kidney region was verified by CPS imaging before administration of the subsequent dose. A 10-s clip was captured at 10 min after each injection to obtain the signal from target-adherent and freely circulating microbubbles in the tumor. A destructive pulse with a high MI of 1.9 was then applied for 2 s to destroy all microbubbles within the ultrasound field. A second 10-s clip, containing signal from the tumor tissue and

circulating microbubbles without the target-bound microbubbles, was recorded immediately after the destructive pulse. All ultrasound images were recorded digitally for later off-line analysis. All imaging procedures in this study were performed by a single ultrasonographer with 5 y of experience.

Image analysis

Contrast intensity was analyzed using Axis Auto Tracking Contrast Quantification (ACQ) software (Siemens), which was run on the ultrasound scanner. The images were reviewed and analyzed in random order by a single investigator who had more than 5 y of experience in CEUS imaging and evaluation. For assessment of contrast enhancement of renal tumors with the three different microbubbles, a region of interest (ROI) encompassing the tumor was manually defined by drawing a free-form contour at the interface of the tumor and surrounding tissue, such that the ROI lay completely within the tumor. For assessment of contrast enhancement with MB_V in renal tumor and the normal renal parenchyma distant from the tumor, the ROIs were set to the same size (mean = 4 mm², range = 2–5 mm²) and placed as close to the same depth as possible in both tumor and normal renal parenchyma, to avoid depth-related effects on image intensity. The quantitative image intensity of the ROI could be obtained with the time-intensity curve. The magnitude of the ultrasound signal caused by target-bound microbubbles alone was calculated by averaging and subtracting the post-destructive pulse signal from initial pre-destructive signal.

Histology and immunohistochemistry

All mice were euthanized immediately after ultrasound imaging, and tumor and adjacent tissues excised. A section of renal parenchyma distant from the tumor was also harvested from each mouse for use as a negative control. The tissues were fixed in 4% paraformaldehyde, paraffin embedded and sliced. The slices were rinsed with phosphate buffered saline for 3 min and blocked with 10% goat serum for 15 min at room temperature. Slices were then incubated with rabbit anti-mouse VEGFR2 polyclonal antibody (bs-0565R, BIOSS Biotechnology, Beijing, China) overnight at 4°C and visualized using horseradish peroxidase-labeled goat anti-rabbit secondary antibody in a 1:200 ratio (ZSGB-BIO, Beijing, China). Representative photographs were taken of randomly chosen fields of view at $\times 200$ magnification.

Statistical methods

The data analysis software SPSS16.0 (SPSS, Chicago, IL, USA) was used for statistical analysis. Data were expressed as means $\pm$ standard deviations. A paired sample *t*-test was performed to compare the contrast

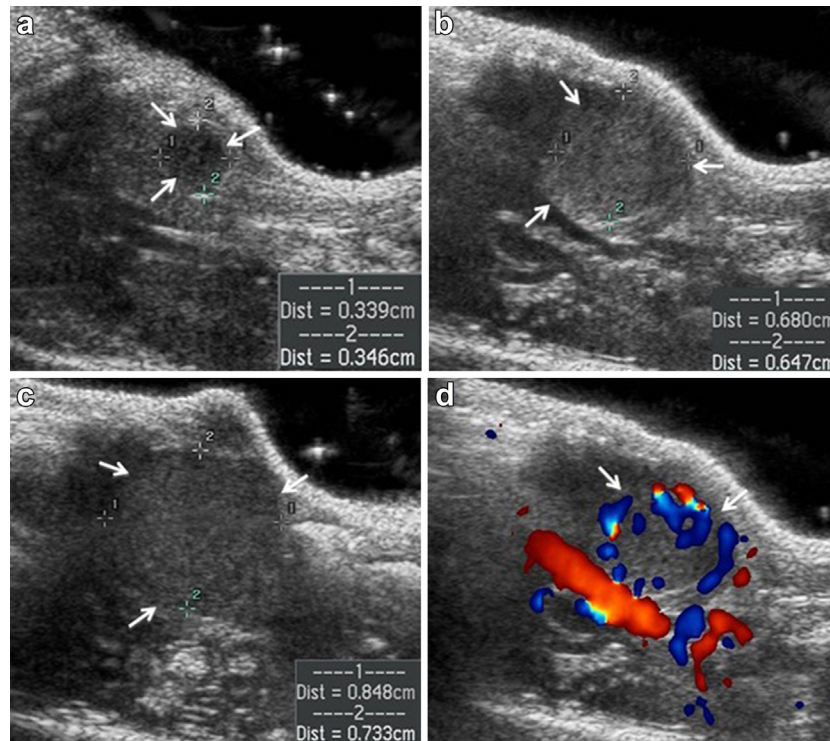


Fig. 1. B-Mode and color Doppler flow imaging of the orthotopic renal carcinoma (arrows) mouse model. (a–c) B-Mode imaging for different-sized tumors. (d) Color Doppler flow imaging of tumor.

intensities attributed to the different types of microbubbles in the same animals, and to compare the contrast intensity in renal tumor with that in normal renal parenchyma. A p -value < 0.05 was considered to indicate statistical significance.

RESULTS

Orthotopic renal tumor mouse model

An orthotopic mouse model of human renal clear cell carcinoma was established by surgical implantation of 786-0 tumor tissues into the kidneys of nude mice. Conventional B-mode ultrasound scanning and color Doppler flow imaging were performed to determine the location, size and macroscopic blood flow of the tumor. All 15 animals developed primary renal tumors. Tumors grew to approximately 5–8 mm by 31 d after the tumor implantation, as determined by B-mode ultrasound imaging. Color Doppler flow imaging revealed a rich vasculature in the orthotopic renal tumor (Fig. 1), suggesting ample blood flow at this time point.

Targeted CEUS imaging of tumor

Microbubble contrast agents targeting VEGFR2 were evaluated *in vivo* 31 d after tumor implantation. Wash-in of each microbubble preparation was assessed during the early (vascular) phase for each animal

(Fig. 2). The mean time to peak intensity within the tumor was 6.32 s (range: 4.66–8.28 s). Peak intensity within the tumor for MB_N, MB_C and MB_V was 28.71 ± 6.14 , 31.22 ± 3.76 and 32.41 ± 3.22 dB, respectively. These values were not significantly different ($p > 0.05$), suggesting that the intravascular behavior was identical for all microbubble preparations within the tumor, as would be expected.

The wash-out phase was analyzed within the tumor region for each microbubble preparation. It was found that the intensity of MB_N and MB_C decayed more quickly than that of MB_V. The duration of higher contrast intensity for MB_V was much longer than that for MB_C and MB_N in the 10-min period after injection (Fig. 3), suggesting persistence of the contrast agent within the vasculature. The mean differences between pre-destruction and post-destruction contrast intensity for MB_V, MB_C and MB_N were 14.55 ± 1.32 , 6.89 ± 1.68 and 5.78 ± 0.88 dB, respectively, MB_V had a significantly higher contrast intensity than MB_C and MB_N ($p < 0.01$) (Fig. 4). This indicates specific accumulation of VEGFR2-targeted microbubbles in the tumor vasculature.

After injection of the microbubbles targeting VEGFR2 (MB_V), enhancement of the tumor was faster than that of adjacent normal renal parenchyma (Fig. 5); the peak intensity of the renal tumor was significantly

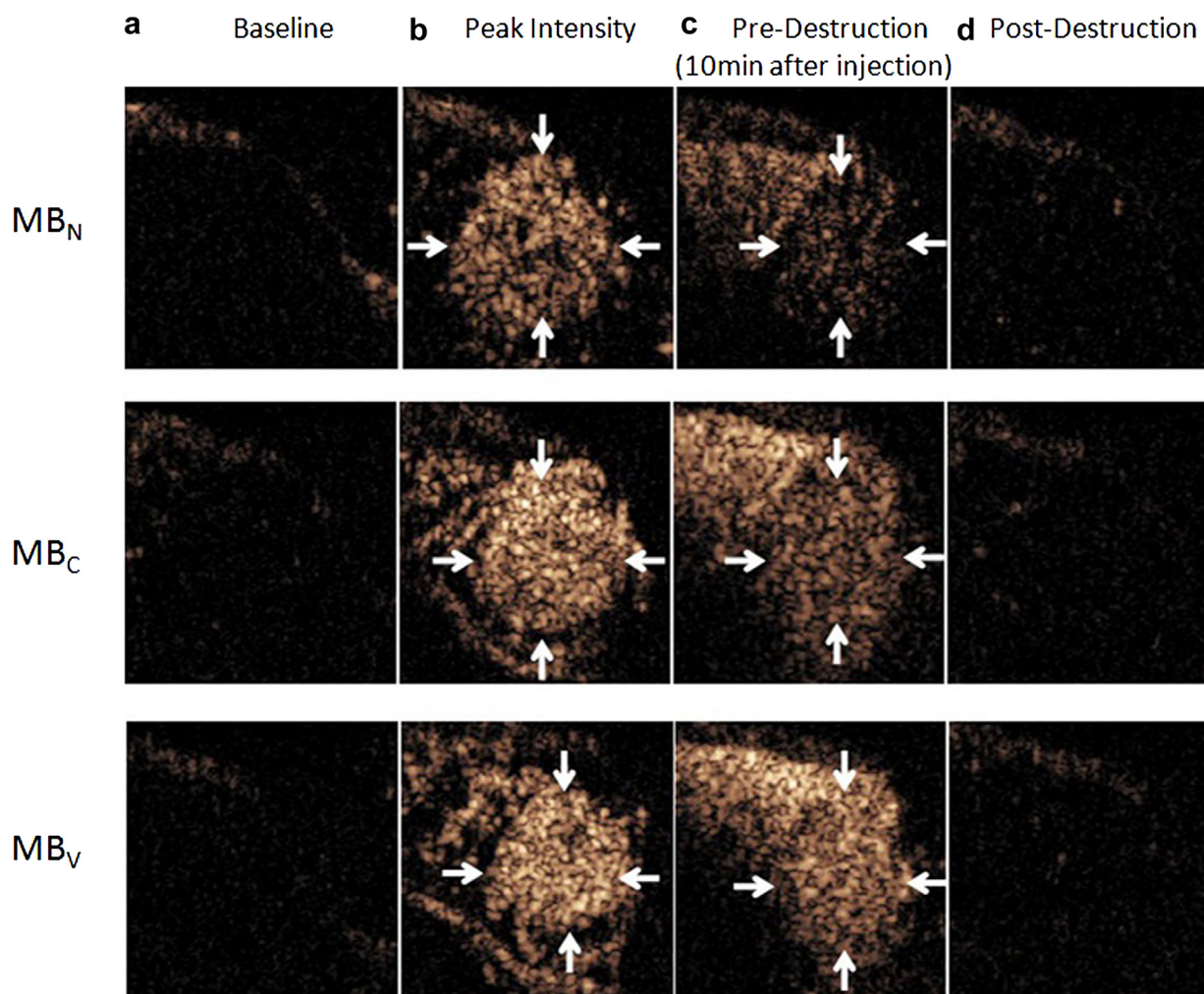


Fig. 2. Contrast-enhanced ultrasound images of orthotopic renal tumor (arrows) with MB_N, MB_C and MB_V in nude mouse. Shown are representative images from a single mouse taken over the study duration. (a) Before microbubble injection (baseline). (b) Peak intensity after injection. (c) Pre-destruction (10 min after injection). (d) Post-destruction. MB_N = naked (unconjugated) microbubbles, MB_C = microbubbles conjugated with IgG antibody (control), MB_V = microbubbles conjugated with vascular endothelial growth factor receptor 2.

higher than that of normal parenchyma (35.17 ± 3.98 dB vs. 31.01 ± 2.04 dB) ($p < 0.05$), suggesting altered perfusion within the tumor relative to healthy tissue. Additionally, wash-out of MB_V from the parenchyma was faster than that of the tumor: at 10 min after injection, almost no MB_V were retained in the normal parenchyma. The baseline-subtracted contrast intensities for renal tumor and normal parenchyma were 13.25 ± 2.90 and 3.62 ± 1.79 dB, respectively. The renal tumor had a significantly higher baseline-subtracted contrast intensity than normal parenchyma ($p < 0.01$) at $t = 10$ min (Fig. 6).

Expression of VEGFR2 in orthotopic renal tumors

Immunohistochemistry staining for VEGFR2 on paraffin sections cut from the orthotopic renal tumor after

ultrasound imaging revealed extensive expression at tumor neovascular endothelial cells. Normal renal parenchyma exhibited negligible staining for VEGFR2 (Fig. 7).

DISCUSSION

Molecular ultrasound imaging with microbubbles targeting molecular markers of angiogenesis was first performed almost 10 y ago (Ellegala et al. 2003; Leong-Poi et al. 2005), and subsequent studies indicated its feasibility in monitoring response to therapy (Korpanty et al. 2007; Lyshchik et al. 2007). There are now several angiogenesis-targeted microbubble formulations made with clinically suitable ligands and conjugation chemistries (Anderson et al. 2010,

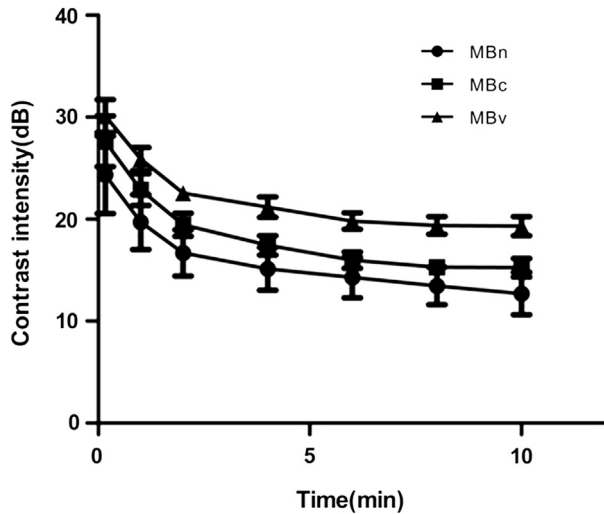


Fig. 3. Time-intensity curve of orthotopic renal tumor with MB_N, MB_C and MB_V in nude mice ($n = 15$) in the wash-out phase. MB_N = naked (unconjugated) microbubbles, MB_C = microbubbles conjugated with IgG antibody (control), MB_V = microbubbles conjugated with vascular endothelial growth factor receptor 2.

2011; Pochon *et al.* 2010; Tardy *et al.* 2010), although none have yet advanced to clinical approval. High spatial resolution and the lack of ionizing radiation are compelling advantages for ultrasound over other imaging modalities.

There is a growing body of literature pertaining to VEGFR2 imaging with microbubbles (both commercially available and other formulations). However, this technique has not yet found routine use either in a research setting (*e.g.*, use as an endpoint in drug development studies) or in clinical practice. One reason for this is lack of information regarding the behavior of these agents in various tumors and models of solid cancer. Most of the previous studies in the field relied on subcutaneous tumor

models; although these are sufficient to show proof-of-concept, they generally are far removed from the clinical context and are increasingly shunned by the cancer research field in favor of orthotopic or spontaneous models. Thus, characterization of the behavior VEGFR2-targeted agents in an advanced cancer model represents an important step in the development of this technique.

In this study, we used an orthotopically implanted renal clear cell carcinoma to model the clinical disease. As a highly aggressive tumor, most renal clear cell carcinomas exhibit a hypervascular nature (Jinzaki *et al.* 2000; Yamashita *et al.* 1992) and have a high degree of VEGFR2 overexpression (Tsuchiya *et al.* 2001). In the present study, immunohistochemical staining confirmed the presence of endothelial VEGFR2 in the 786-0 xenograft tumors, with minimal expression in adjacent kidney tissue. The orthotopic model has the ability to reflect inherent invasive and metastatic behavior of this tumor, as its environment is similar to that of the originating organ (Fidler *et al.* 1990). In the context of research, our study indicates the feasibility of ultrasound molecular imaging as a truly non-invasive method for assessing the molecular phenotype of orthotopic tumors, in this case within the kidney. This may offer both economic and scientific benefit to small animal researchers, as molecular imaging may offer more meaningful information than caliper measurements and may enable the use of fewer mice per study.

In this study, we used a clinical ultrasound scanner operating at 7–14 MHz. Imaging of small animals with a clinical ultrasound scanner can be challenging because of its relatively lower spatial resolution compared with ultra-high-frequency ultrasound scanners. As the mouse kidney is a relatively well-defined organ, the spatial resolution offered at 7 MHz was adequate for identifying the organ and localizing the tumor within the kidney.

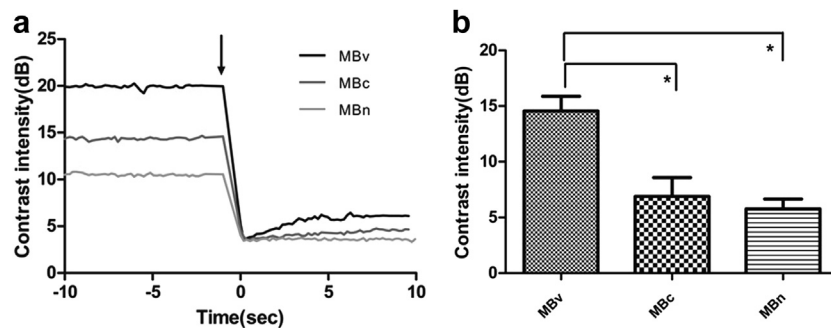


Fig. 4. Difference between pre- and post-destruction contrast intensity. (a) Representative pre-destruction and post-destruction contrast intensity of MB_N, MB_C and MB_V in a single mouse. The temporal location of the microbubble destruction sequence is indicated by the arrow. (b) Subtracted mean pre-destruction and post-destruction contrast intensity for MB_N, MB_C and MB_V ($n = 15$) ($*p < 0.01$). MB_N = naked (unconjugated) microbubbles, MB_C = microbubbles conjugated with IgG antibody (control), MB_V = microbubbles conjugated with vascular endothelial growth factor receptor 2.

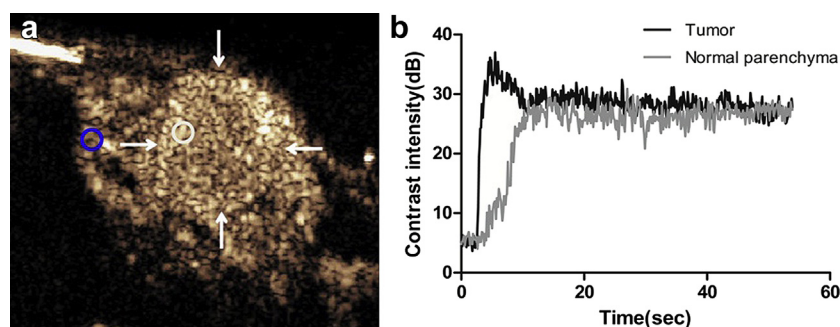


Fig. 5. Enhancement of orthotopic renal tumor and normal renal parenchyma after injection of MB_V (microbubbles conjugated with vascular endothelial growth factor receptor 2). (a) Peak enhancement of orthotopic renal tumor (*arrows*) in contrast-enhanced ultrasound imaging. (b) Time-intensity curve of orthotopic renal tumor (relative to white region of interest) and normal renal parenchyma (relative to blue region of interest).

Clinical scanners operating at lower frequencies (such as the 7 MHz used here) generally offer greater sensitivity to microbubbles than scanners operating at ultra-high frequency (18–40 MHz). For molecular imaging, where sensitive detection of microbubbles is critical, we found that a clinical scanner provided the greatest utility in our mouse model. We were able to locate tumors as small as 1 mm using B-mode imaging. However, for

applications in which very high spatial resolution is paramount (*e.g.*, in sub-millimeter tumors or for precisely identifying the location of tumors), imaging at very high frequencies may be preferred.

Our study indicated that targeted microbubbles can be used to identify VEGFR2-expressing tumor within the kidney. The contrast intensity within the tumor region was significantly higher for VEGFR2-targeted

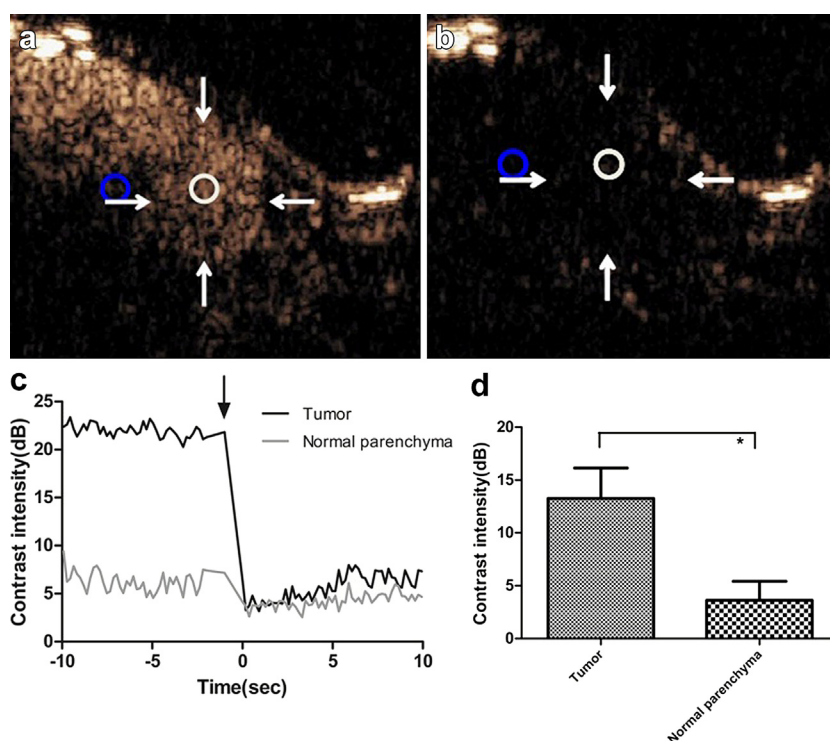


Fig. 6. Difference between pre- and post-destruction contrast intensity after injection of MB_V (microbubbles conjugated with vascular endothelial growth factor receptor 2) in orthotopic renal tumor (*arrows*) versus normal parenchyma. (a) Pre-destruction (10 min after injection). (b) Post-destruction. (c) Time-intensity curve of pre- and post-destruction contrast intensity of orthotopic renal tumor (relative to white region of interest) and normal parenchyma (relative to blue region of interest) in a single mouse. The temporal location of the microbubble destruction sequence is indicated by the *arrow*. (d) Subtracted mean pre-destruction and post-destruction contrast intensity for orthotopic renal tumor and normal parenchyma ($n = 15$) (* $p < 0.01$).

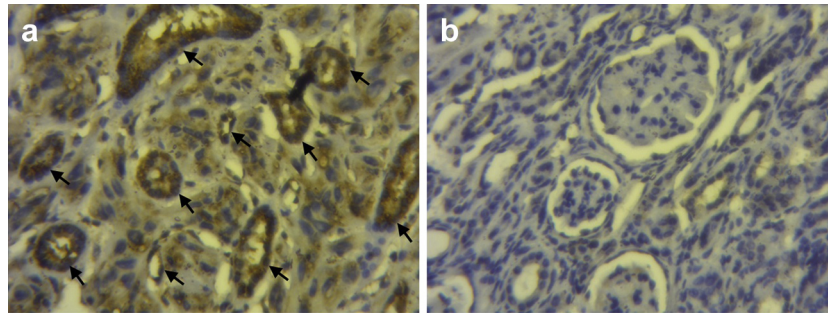


Fig. 7. Immunocytochemistry staining of renal clear cell carcinoma 786-0 orthotopic xenograft tumor and normal renal parenchyma with anti-vascular endothelial growth factor receptor 2 (VEGFR2) antibody (200 × magnification). (a) Tumor tissue (regions of VEGFR2 expression are indicated by *arrows*). (b) Normal renal parenchyma.

microbubbles relative to microbubbles bearing a control (isotype) ligand or non-targeted (no ligand) microbubbles. Additionally, the VEGFR2-targeted microbubbles exhibited delayed enhancement within the tumor, indicating retention. One potential difficulty in our model is the proximity of the kidney to the liver. The liver plays a major role in clearance of circulating microbubbles (Tartis *et al.* 2008; Willmann *et al.* 2008; Yanagisawa *et al.* 2007), and significant non-specific retention was observed. For example, the liver is located in the upper left region of the image in Figure 2. Thus, it is critical to use B-mode imaging to first establish organ orientation in advance of performing the contrast scan.

Some studies have reported that the increased video intensity after intravenous injection of targeted microbubbles was caused primarily by the circulating microbubbles in tumor vessels in the initial frames and by attachment of microbubbles to tumor vessel endothelial cells in the later frames (Jun *et al.* 2010). Our study also indicated that as an orthotopic renal tumor model, the tumor had significantly higher contrast intensity than normal renal parenchyma, and immunohistochemistry confirmed that there was higher VEGFR2 expression in renal tumor than in the normal renal parenchyma, which was consistent with the literature (Tsuchiya *et al.* 2001).

Based on its pivotal role in tumor growth and metastasis, VEGFR2 is a versatile target for anti-angiogenic drugs (Folkman 2006). It has been suggested that the efficacy of VEGF/VEGFR2-directed therapies does not necessarily depend on dosage, but rather on the amount of VEGF/VEGFR2 expression in the tumor micro-environment (Cai *et al.* 2006). Thus, methods for visualizing and quantifying VEGFR2 expression may be important in various anti-cancer treatments. A potential strength of molecular imaging techniques is the ability to assess the expression of drug targets throughout the organ, thereby avoiding the sampling artifacts inherent in biopsies. As contrast-enhanced ultrasound imaging with

microbubbles targeted to VEGFR2 enables the spatial visualization and quantitative assessment of the molecular expression of VEGFR2 non-invasively and with high resolution, it may greatly help in optimizing anti-cancer regimens with respect to choice of the appropriate anti-angiogenic drug dose and time window. In addition to its clinical potential, this technique may be beneficial to preclinical research in the drug discovery field. It should be noted that the microbubbles employed in this study (Targestar-SA) are not suitable for clinical use because of the streptavidin component, although as noted above, several agents have been developed to be translatable to clinical use. These agents use ligands reactive in humans and do not require point-of-use ligand conjugation and washing. Our study is a preliminary experiment in ultrasound molecular imaging of renal cancer. Further studies are needed to investigate the optimal doses and parameters for contrast-enhanced ultrasound imaging with VEGFR2-targeted microbubbles in this renal tumor model.

There are several limitations to our study that should be considered. Microbubble agents were administered as a bolus in 10 μ L. Injection of such a small volume introduces some uncertainty in the measurement of dose. However, this uncertainty is counterbalanced by the need to avoid volume overload, which may be a concern when performing multiple doses in a single study. Microbubble formulations were washed to remove any unconjugated antibody. This procedure results in reduction of microbubble concentration and may introduce uncertainty into the final microbubble concentration. To control for this, all microbubble samples were conjugated and washed in parallel using the same procedure, and injected in random order. Only a single 2-D field of view was imaged in each animal, rather than the full kidney volume. Use of newly developed volumetric imaging procedures (Hu *et al.* 2013; Streeter *et al.* 2013) may result in improved sensitivity. Volumetric imaging may also make feasible correlation of the observed imaging signal with

the true target density. Finally, our study indicated that VEGFR2-targeted microbubbles can differentiate between tumor and parenchymal tissue in an orthotopic renal tumor, but did not determine whether this technique can detect potentially subtle changes in VEGFR2 expression in response to therapy; subsequent studies are needed for this determination.

CONCLUSIONS

The behavior of microbubbles targeting VEGFR2 was evaluated in an orthotopic mouse model of renal cell carcinoma. VEGFR2-targeted microbubbles were found to have significantly higher contrast intensity in the late phase than sham-targeted or naked microbubbles. This suggests that this technique may be useful as a non-invasive endpoint in preclinical research and, eventually, as a diagnostic in clinical medicine.

Acknowledgments—This work was supported by a grant from the National Natural Science Foundation of China (No. 30970835) to B.Y.

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